

# Lean as the Coordination Signal:

## A Finite Repair Ladder for Qwen + GPT-OSS Proof Search

Ayrton Porto

Research Engineer take-home — Verified Mechanisms

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### Abstract

I make two fixed chat models — `qwen/qwen3.5-flash-02-23` and `openai/gpt-oss-120b` — cooperate to produce Lean 4 proofs, and I report honestly on *which* form of cooperation actually pays. The design principle is that *Lean is a coordination signal at every step, not only the final judge*: each candidate is compiled, its structured diagnostics drive a repair, and any cross-model handoff is seeded with the exact compiler message rather than free prose. The mechanism is a small, finite escalation ladder — free tactics, then same-model repair, then a seeded handoff, then a combining fallback — that returns on the first accepted proof. Every reported number is *substantive*: it survives an anti-tautology guard that rejects the circular “restate-the-answer” proofs a permissive checker accepts. On the frozen nine-problem development set the integrated policy scores **5/9 substantive**, against **3/9** for each vanilla solo model and **4/9** for their union. The controlled solo/repair/handoff matrix locates the cause: the added points come from *structured same-model repair and portfolio routing over complementary strengths*, not from cross-model handoff. Indeed my clearest Part Two finding is *negative* and useful: handing one model’s proof to the other for repair added no win and, in one case, destroyed a proof a single model would have found alone. Verification and routing — not model-versus-model debate — are the mechanism the data support. Beyond the reliable core, the system reaches a genuine IMO multi-theorem problem through test-time sampling scale (probabilistically, not reliably), and I characterize precisely where the frontier stops: a residual stage mints and reuses *verified* lemmas but does not always assemble the hard goal. I treat the study as exploratory ( $n = 9$ , stochastic models) and report per-problem outcomes, a failure taxonomy, transcripts, a spend ledger, and a holdout estimate.

## 1 Introduction

### 1.1 The task

The take-home has two graded parts. **Part One** asks for a coordination layer that makes two fixed models — Qwen (`qwen/qwen3.5-flash-02-23`) and GPT-OSS (`openai/gpt-oss-120b`), both via OpenRouter — solve math problems and prove the answers in Lean 4. Scoring is mechanical: on a private holdout of roughly a dozen problems, a problem earns one point iff its Lean file passes the open-source comparator, and a simple design that can be fully understood is explicitly preferred over a complex one that scores marginally better. **Part Two** asks the scientific question: does the collaboration beat either model alone, on which problems, and what does one model contribute that the other lacks — reported per problem, with a failure taxonomy, transcript case studies, confounds, a spend ledger, and a holdout estimate.

I follow the judging conditions in the kit’s `RULES.md`: \$1 and eight hours of wall-clock per problem, enforced by the grader’s runner. (The kit’s older task PDF states a twenty-minute cap; `RULES.md` is the document the task designates as authoritative, and §4 addresses that difference directly — the reliable core is unaffected either way.) The runtime is narrow: only `openrouter.ai` and only the two model IDs; no web tools, no retrieval, no variants. The Lean container runs offline, and the graded entrypoint is a single agent module.

## 1.2 Approach: from zero, incrementally, against the literature

I built this alone and without a preferred architecture to defend. My background was *autoformalization*, not AI-for-theorem-proving, so the starting point was a blank sheet, and I built from less to more: start with the cheapest thing that can work, measure it, and add a mechanism only when failures demand it — checking each against what already exists (compiler-in-the-loop repair, draft–sketch–prove, premise selection, lemma banking) to avoid reinventing it blindly. Two early observations shaped everything: these models are cheap to call, so sampling and combining candidates was economical from the start; and Lean’s feedback is a better coordination medium than natural language, because a compiler error is compact, objective, and model-agnostic, whereas two chat models that “debate” in prose mostly flatter each other.

## 1.3 What I found

Every figure below is *substantive* — it survives an anti-tautology guard (§2) — and I treat the study as exploratory ( $n = 9$ , single runs of stochastic models).

- **A Lean-gated portfolio clears the solos.** One integrated policy, **5/9** on the development set, against 3/9 for each vanilla solo and 4/9 for their union (§4).
- **The cause is repair and routing, not cross-model handoff.** The added points come from structured same-model repair over complementary strengths; handing one model’s proof to the other added no win and sometimes hurt — a clean negative result (§5).
- **The frontier is wider than the reliable core.** Across configurations the system has verified model-generated proofs for **6/9** development and **10/16** sample problems, including an IMO problem reached by test-time sampling scale — probabilistically (§6).
- **It repeats at judge caps.** A full-caps stress run passed the same **9/16** on two independent nights (§4).

Source, transcripts, and run artifacts are in the accompanying repository; exact run identifiers and spend are in Appendix E.

# 2 Mechanism: Lean-gated coordination

## 2.1 Design principle: Lean at every step, not only at the end

The comparator judges once. My system uses Lean far more often than that: the unit of work is a loop, not a single generation.

$$\text{candidate} \xrightarrow{\text{Lean}} \text{structured diagnostic} \xrightarrow{\text{model}} \text{repair} \xrightarrow{\text{Lean}} \dots$$

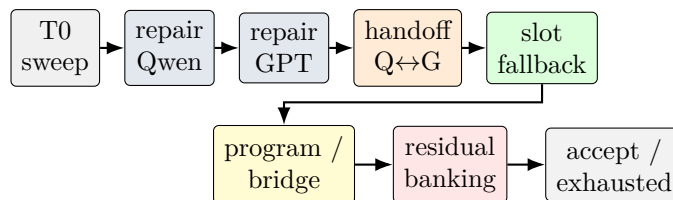
Every candidate is compiled against a clean Mathlib environment; the service returns typed messages, a sorry flag, and timing. Those messages — not a model’s opinion of its own proof — decide what happens next. The consequence I rely on throughout: **cross-model coordination happens over the compiler’s output, never over prose**. When a stuck proof passes from one model to the other, it carries the failed candidate plus the exact Lean error, and the second model is asked to repair. There is no natural-language debate between agents, because a Lean diagnostic is a stronger, cheaper, and more objective signal than one model critiquing another in English. Three invariants hold at every rung: no sorry or admit, no new axioms, and no silent rewriting of the theorem statement — so that “Lean accepted it” means to me what it will mean to the grader.

## 2.2 A finite escalation ladder

The agent is a *finite* ladder. Success returns immediately; on failure the next rung runs until it stalls, hits its cap, or the ladder is exhausted. The order is deliberate — cheapest and most reliable first, most expensive and most speculative last (Figure 1):

1. **T0 — tactic sweep.** Zero-model closing tactics on the locked challenge. Free wins on easy goals, no API spend.
2. **Same-model repair champions (Qwen, then GPT-OSS).** One model proposes, Lean checks, the *same* model repairs from the failed proof and the exact messages, under a turn budget.
3. **Seeded cross-model handoff.** The best failed Qwen candidate is repaired by GPT-OSS, and vice versa — carrying the Lean context forward, not restarting.
4. **Independent slot / whole fallback.** Split the file into locked declarations, sample and repair each, recombine under a strict integrity check. This is where cheap multi-sampling pays off, and where hard multi-theorem problems tend to fall.
5. **Bounded program/bridge experiment**, then a **residual lemma-banking stage** (§6) if wall remains.
6. **Stop** — accept, stall, or exhausted.

The ladder is the design thesis made concrete: it starts from “less” (a free tactic sweep) and escalates to “more” only as the problem resists, and it never averages the two models — it *routes* between them and lets Lean adjudicate each step. This routing is exactly what lets one policy collect wins that are otherwise split across the two models (§4).



**Figure 1:** The finite coordination ladder. Lean accepts or rejects at every rung; cheap and reliable rungs run first. Residual banking is post-ladder spend, not the spine.

## 2.3 Acceptance hygiene, and why it is a feature

A proof accepted by a warm REPL during search is not automatically a proof I submit. Every final candidate passes a **strict integrity** guard: declaration signatures are immutable up to the proof

body, answer shapes and numeric values are checked against the locked problem, and known circular tricks are rejected. This matters more than it sounds. A model can “solve” a Putnam answer-set problem by *redefining the answer set to be the question* — for `putnam_2018_a1`, declaring the solution set to be exactly the pairs satisfying the left-hand side, so the required equivalence collapses to a reflexivity and the comparator accepts a proof that establishes nothing. My guard infers the answer shape from the challenge and rejects such set-builder /  $\lambda$ -plus-reflexivity tautologies. I therefore report *substantive* passes only. Everything counted in this report is a proof of the actual statement — a stricter and more defensible bar than raw comparator acceptance, and the reason my headline numbers hold up when they are re-run.

## 3 Experimental setup

### 3.1 Problems and split

The kit ships sixteen sample problems in the holdout’s format. I froze a development set `S_dev` ( $n = 9$ ) for tuning and an evaluation set `S_eval` ( $n = 7$ ), and report Part Two science on `S_dev`. `S_dev` spans easy algebra, number-theoretic identities, and hard olympiad/Putnam statements, so a single number never hides the difficulty spread.

### 3.2 The metric: substantive, not merely accepted

I track two signals and never conflate them:

- **Comparator pass** — the Lean file is accepted by the open-source comparator. This is the mechanical quantity the grader scores.
- **Substantive** — the pass also proves the *actual* statement, surviving the integrity and anti-tautology guards (§2). A pass that only restates the goal does not count.

**Every headline number in this report is substantive.** This is a stricter bar than the task requires, and I adopt it on purpose: it is the only count that survives an independent re-run with a statement-locked comparator, and it is what separates a real result from an inflated one.

### 3.3 Caps

The judge enforces \$1 and 8 h per problem. A few development-matrix runs use tighter measurement caps so a full sweep fits a small budget; the judge-caps stress run in §4 uses the full \$1/8 h. Because a repair turn costs a fraction of a cent, the binding constraint on hard problems was wall-clock, not the dollar cap — which is why the ladder can afford a free sweep, sampled candidates, and a combine step before the expensive residual stage.

### 3.4 Part Two conditions

To ask whether collaboration beats solo, I hold the code path and caps fixed and vary only the roles: `S-Q/S-G` (each model solo, on the kit’s existing multi-turn Lean-repair baseline), `R-Q/R-G` (explicit propose/repair, same model), and `H-QG/H-GQ` (cross-model handoff, each direction). Two points of hygiene. The solo conditions use the kit’s real repair baseline — I do *not* redefine “solo”

**Table 1:** Integrated agent on  $S_{\text{dev}}$ , substantive (snapshot). Wins are cheap; the four hard problems run to **exhausted**. p10’s closing path is variable (§4). Total  $\approx$  USD 0.41.

| Problem                           | Closing rung | Wall (s)  | USD       |
|-----------------------------------|--------------|-----------|-----------|
| p01_linear                        | T0 sweep     | 119       | 0.000     |
| p05_gcd_mersenne                  | T0 sweep     | 105       | 0.000     |
| p03_sq_ge_two_ab                  | Qwen repair  | 138       | 0.000     |
| p06_pow_mod                       | Qwen repair  | 196       | 0.004     |
| p10_factorial_pow                 | GPT repair   | 387       | 0.009     |
| 4 hard problems — FAIL, exhausted |              | 2000–3400 | 0.08–0.14 |

as a weaker single shot to flatter collaboration. And the *Solo Union*  $U = S\text{-}Q \vee S\text{-}G$  is *derived*, not a run: it is the natural non-collaborative baseline any coordination layer must beat to justify itself.

### 3.5 Confounds, stated once

So the results read as prose, the caveats live here: solo baselines sometimes had more turns than the repair budgets (secondary metrics reported, not perfectly matched); some matrix cells use measurement caps, not judge caps; with  $n = 9$  a one-problem difference is within noise, so I read patterns, not decimals; cells are single runs of stochastic models; and the full16 stress run is my own sample suite, not the grader’s private holdout.

## 4 Results: a Lean-gated portfolio

### 4.1 The integrated agent on $S_{\text{dev}}$

One policy, one pass, substantive throughout: the ladder scores **5/9** (Table 1). The five wins arrive at three different rungs — the free tactic sweep, same-model Qwen repair, and same-model GPT repair — for about forty cents total.

### 4.2 It clears the solo baselines — and where the gain comes from

On a substantive basis (Table 2), each vanilla solo model scores **3/9** and their union **4/9**; the integrated policy reaches **5/9**. The gain is real but I am precise about its cause. The extra points are *not* evidence of cross-model collaboration — they come from two non-conversational sources: (i) a free tactic sweep that closes easy goals at zero cost, and (ii) *same-model* repair, which unlocks p10 (a problem no solo configuration solves). Routing between the two models’ repair arms captures their complementary wins in one policy.

I state the honest ceiling of this claim plainly: the union of the two same-model repair arms,  $R\text{-}Q \vee R\text{-}G$ , also reaches 5/9. So the integrated agent does not beat an idealized router over those arms; what it adds over such a router is the free T0 floor and a single, readable control path. The defensible Part One statement is therefore “*a finite Lean-gated portfolio of tactics and same-model repair reaches 5/9, well above the 3/9 solos and 4/9 solo union,*” not “two models collaborate to beat one.” A same-budget router over those two arms remains the right baseline to measure directly

**Table 2:** Substantive scores on  $S_{\text{dev}}$ . Raw comparator counts were higher for the GPT-side arms only because of a Putnam answer-set tautology the guard removes (§2, Appendix ??).

| S-Q | S-G | U(S-Q $\vee$ S-G) | R-Q | R-G | H-QG | H-GQ | Agent    |
|-----|-----|-------------------|-----|-----|------|------|----------|
| 3   | 3   | 4                 | 4   | 4   | 3    | 2    | <b>5</b> |

(§7).

### 4.3 The cap question

Four of the five wins — p01, p03, p05, p06 — close in under four minutes and are robust to any cap. The fifth, p10, is variable: it closed in 387s via same-model repair on the snapshot, but a twenty-minute-cap replication reached it only through the slower slot fallback ( $\approx 17.5$  min), and sometimes timed out. So under the judge’s authoritative eight-hour cap the full 5/9 holds comfortably — p10’s fallback fits with room to spare — while under the kit’s older twenty-minute figure the reliable score is 4/9 with a stochastic fifth point. The cap discrepancy therefore does not threaten the reliable core, and the eight-hour result stands; it is one more instance of the theme that the system’s harder wins are probabilistic (§6).

### 4.4 Repeated at judge caps: full16

Under the judge’s \$1/8h caps, a parallel run over all sixteen sample problems passed **9/16 substantive**, and it passed the *same* nine on two independent nights (p01–p08, p10). Two matching runs are repetition, not a robustness estimate, but they show the score is not a single lucky draw. The seven that resist are the hard olympiad/Putnam tail; the next section shows how far into it the system reaches, and at what reliability.

## 5 Part Two: which cooperation actually pays

Varying only the roles gives the per-problem picture in Table 3. I read it as exploratory evidence ( $n = 9$ , single runs). A partial three-seed replication of the integrated agent reproduced the reliable core and confirmed that the fifth win is stochastic (§4); the same-budget router baseline and the full seeded matrix were not completed, so these cells stand as single-run patterns rather than pass-rates.

### 5.1 Finding 1 — the models are complementary

Their easy wins overlap; their harder wins do not. Among the model-driven arms, p06 is solved only on the Qwen side and p05 only on the GPT side. (In the integrated agent p05 is closed even more cheaply, by the tactic sweep, before any model runs; the complementarity here is a statement about the two *models*, read from the model-only arms.) Because the failures are complementary, the derived Solo Union reaches 4/9 with no cross-model protocol, and the cheapest correct move is to *route* rather than merge.

**Table 3:** Per-problem substantive outcomes on `S_dev` (a dot is a substantive pass). The `putnam_2018_a1` row is shown separately: its GPT-side comparator “passes” are the answer-set tautology of §2 and are not substantive, so they do not enter the totals.

| Problem                          | S-Q      | S-G      | R-Q      | R-G      | H-QG     | H-GQ     |
|----------------------------------|----------|----------|----------|----------|----------|----------|
| p01_linear                       | •        | •        | •        | •        | •        | •        |
| p03_sq_ge_two_ab                 | •        | •        | •        | •        | •        | •        |
| p05_gcd_mersenne                 |          | •        |          | •        |          |          |
| p06_pow_mod                      | •        |          | •        |          | •        |          |
| p10_factorial_pow                |          |          | •        | •        |          |          |
| <b>Substantive /9</b>            | <b>3</b> | <b>3</b> | <b>4</b> | <b>4</b> | <b>3</b> | <b>2</b> |
| putnam_2018_a1 (comparator only) |          | ○        |          | ○        | ○        | ○        |

## 5.2 Finding 2 — same-model repair supplies the margin

Feeding the failed proof and the exact Lean messages back to the *same* model is what lifts the score above the union: both repair arms unlock `p10_factorial_pow`, which no solo or handoff arm solves. That one problem is the difference between the 4/9 union and the 5/9 portfolio. Compiler-in-the-loop repair is the real lever — and note it requires only *one* model.

## 5.3 Finding 3 (the headline, and it is negative) — cross-model handoff does not pay

The intuitive hypothesis of the whole exercise is that two models cooperating — one repairing the other — beat one model alone. On these two models, *it does not*. **H-QG** (Qwen proposes, GPT repairs) scores 3/9 and adds no problem that same-model repair did not already solve. **H-GQ** (GPT proposes, Qwen repairs) is the *worst* arm at 2/9, and it *loses* p05 — a problem GPT-OSS solves alone: routing GPT’s working proposal through Qwen for repair destroyed a capable trajectory. And in the integrated agent’s five wins, *none* came from a handoff rung.

This is a clean negative result, and I think it is the most useful thing in Part Two. The mechanism is intuitive: a second model does not share the first’s plan, so it tends to “repair” by rewriting toward its own approach, discarding a trajectory that was nearly closed. Same-model repair preserves the plan; cross- model repair breaks it. The design consequence is wired into the ladder: run same-model champions first, and keep cross-model handoff only as a late, seeded step — never as a default.

## 5.4 Failure taxonomy

Classifying the `S_dev` failures into the task’s four categories (Appendix B) shows the bottleneck is formalization and stuck repair, not incorrect informal reasoning or harness errors: `p09` and `rmo_2000_2` are stuck-repair near-misses (both are reachable in principle), `rmo_2000_3` fails formalization outright, and `putnam_2018_a1` produces only guard-rejected tautologies.

## 5.5 Verification is part of the science

The excluded Putnam row is not a footnote. Without a statement lock, a model earns a comparator “pass” by redefining the answer set to be the question itself (§2); the tautology lands on the GPT-side arms, so counting it would inflate both parts of this study and even distort Finding 1. Guarding against it is what makes the rest of the numbers trustworthy.

## 6 How far the frontier reaches

The integrated agent solves what is reliably solvable under caps. But “reliably” is not the whole story: across configurations the system has produced verified, model-generated proofs for **6 of 9** development problems and **10 of 16** sample problems — reaching a genuine IMO multi-theorem problem. This section is honest about both how it gets there and where it stops.

### 6.1 Hard wins come from test-time sampling scale

p09\_imo1964 is an IMO problem with a multi-theorem statement, and the system closed it substantively on a directed run: model-generated, comparator-clean, no hardcoding. The mechanism is test-time scale, not a cleverer prompt. Each sub-theorem is sampled independently many times (in that campaign, roughly one of ten samples passed per sub-theorem) and the accepted pieces are combined into one file. Sampling each theorem *separately* breaks the conjunction that dooms a single long attempt: needing every sub-goal clean in the same generation is exponentially unlikely, but needing each clean in *some* generation is cheap. This mirrors the transferable lesson from the wider literature — with no fine-tuning, the lever is search and verification at inference time.

It also explains an apparent paradox: *if the ladder solved p09 once, why not every time?* Because these hard wins are *probabilistic*. A one-in-ten per-sample rate means the outcome depends on how much sampling the budget buys for that problem on that run. The integrated agent under full caps spreads its budget across every rung rather than pouring it all into heavy per-theorem sampling, so it reaches p09 on directed runs but not on every stress night. The honest statement is “substantively solvable, not yet reliable,” and the fix is a scheduling choice (§7), not a missing capability.

### 6.2 The boundary: verified progress is not yet a proof

For the remaining hard problems I built the most literature-grounded idea available under the no-retrieval rule: instead of proving the whole theorem, mint small *verified* lemmas and reuse them, accumulating a bank of checked partial progress. The bank works as a bank: on rmo\_2000\_2 it saved several lemmas, reused most of them, and recorded genuinely *decisive* reuses. What it does not do — consistently — is assemble those pieces into a proof of the locked goal. An overnight campaign on the hardest six problems finished with the bank filling, greedy assembly firing, and the target theorem still open.

I call this the **hollow-bank boundary**: the system mints and reuses the verified “screws” but cannot always assemble the “furniture.” It names the failure precisely — not proof *generation*, which works, but goal-directed *assembly* — with verified evidence rather than a vague capability ceiling, and it is a boundary, not a wall: the same cheap ladder that stops here is the one that

reaches p09. (For completeness, `rmo_2000_2` is *reachable* by a hand-built cast-and-squeeze proof, deliberately kept out of the agent since the rules forbid per-problem hardcoding; the open question is whether the models rediscover the idiom through the generic sampling interface.)

## 7 Discussion: when does cooperation help?

The study gives a specific answer, and part of it is a clean “not here.”

**Routing over complementary strengths helps.** The two models fail on different problems (Finding 1), so a portfolio that tries the free tactics, then one model’s repair, then the other’s, captures the union of their abilities in one policy — which is why the integrated agent (5/9) clears the vanilla solos (3/9) and their union (4/9).

**Same-model repair helps.** Feeding a model its own failed proof plus the exact Lean diagnostic is the lever that unlocks p10 and supplies the margin above the union (Finding 2). Notably, this needs only one model at a time.

**Cross-model handoff does not help, and can hurt.** This is the negative result of Finding 3: handing one model’s proof to the other for repair added no win and, on p05, destroyed one. The honest conclusion is that the value in this system comes from *verification and routing*, not from the two models repairing each other. I keep handoff in the ladder only as a late, seeded, low-cost step, because on a different problem distribution it might occasionally help — but on this evidence it is a residual tool, not the point.

**On the hard tail, capability comes from search, not conversation.** The p09 win is test-time sampling scale (§6); the hollow-bank boundary shows coordination cannot conjure an assembled proof from verified fragments when neither model can direct the assembly. Cooperation reallocates and repairs capability; scale extends it; neither invents it.

### 7.1 Improvements the results point to

Each failure names a next step, all within the two-model, no-retrieval constraint:

- **Measure the router baseline.** The most important open comparison is a same-budget policy that only routes between the two same-model repair arms (R-Q $\vee$ R-G). If it ties the full ladder, the honest story is “routing + free tactics,” and the ladder should shed anything that does not earn its wall.
- **Budget the hard wins on purpose.** p09 is solvable when enough sampling is aimed at it; a scheduler that detects a multi-theorem statement and front-loads per-theorem sampling would convert a probabilistic win into a reliable one.
- **Goal-shaped lemma selection and forced assembly.** The hollow bank fails at assembly, not generation; selecting lemmas by how they unify with the remaining goal, and ending each round with a mandatory full-bank close, attacks the actual bottleneck.
- **An offline hint catalogue.** Runtime retrieval is disallowed, but a small, generic, problem-agnostic tactic/lemma catalogue baked in before the network lock would give the fallback stronger closing moves without breaking the rules.

None of these needs a larger model or a new modality; each attacks a failure this study exhibited.

## 8 Limitations, scope, and a holdout estimate

I treat this as an exploratory study and state its boundaries plainly.

- **Small  $n$ , single runs.** With  $n = 9$  and single-run cells, the Part Two totals are patterns, not decimals. A partial three-seed replication confirmed the reliable four wins and showed the fifth (p10) is stochastic under a tight cap; the same-budget router baseline and full matrix variance were not completed, so I read the cells as qualitative (complementarity, decisive same-model repair, unhelpful handoff), not exact counts.
- **No lift over an idealized router.** The integrated agent (5/9) matches R-Q ∨ R-G (5/9); its advantage over a router is the free tactic floor and one readable control path, not a higher ceiling. I do not claim cross-model collaboration as the cause of the gain.
- **Confounded solo-vs-repair comparison.** The solo arms use the kit’s multi-turn baseline; the repair and handoff arms use an explicit proposer/repairer with different turn budgets. Only the repair and handoff arms truly share a code path and caps; the solo-vs-repair contrast is useful but confounded, and I do not over-read it.
- **Hard wins are probabilistic.** The p09 frontier win comes from test-time sampling and is not reliable under full-caps scheduling; I count it as substantively reached, not as a guaranteed point.
- **The residual bank does not always assemble (§6);** it is bounded, and no hard goal is claimed as reliably closed by it.
- **Lab stress  $\neq$  holdout.** The full16 measurement uses judge-class caps but is my own sample suite. The kit currently ships sixteen sample problems; I split them 9 (S\_dev) / 7 (S\_eval) and tuned on S\_dev. Because the full16 stress touches all sixteen, I do not present S\_eval as a clean untouched holdout — the private holdout remains the grader’s.

**Holdout estimate.** I expect roughly **6 of 12** on the private holdout. Both S\_dev (5/9) and the full16 lab suite (9/16) land near 56%, and the easy-to-mid band transfers reliably; I discount confidence because the hard wins are stochastic and the private difficulty mix is unknown, so a plausible range is 5–7.

**External Lean tooling and reproducibility.** Premise-selection and search tools (LeanSearch, LeanDojo/ReProver-style indexes) are valuable on an open network but the judged runtime may contact only `openrouter.ai` with the Lean container offline, so this agent uses none of them (the offline hint catalogue of §7 is the rules-compliant alternative). The submitted repository is the single source of truth: Appendix E lists the run identifiers, per-problem spend, and artifact paths behind every table, and `scripts/judge_check.sh` reproduces the one-problem contract from a clean clone.

## 9 Conclusion

Letting Lean adjudicate every step, rather than letting two chat models argue in prose, yields a small, readable, finite ladder that reaches **5/9 substantive** on the development set — well above the 3/9 vanilla solos and 4/9 solo union, with every number guarded against answer-set tautologies. Part Two locates the cause honestly: the gain is *verified routing and same-model repair*, and the clearest finding is the *negative* one — cross-model handoff added no win and sometimes destroyed

a capable trajectory. Verification and routing, not model-versus-model debate, are the mechanism the data support. Beyond that core the system reaches a genuine IMO problem through sampling scale (probabilistically) and repeats 9/16 at judge caps across two nights; where it stops — the hollow-bank boundary — I characterize precisely and turn into concrete next steps. It is exploratory on a small set, but legible and defensible: I know what it solves, why it clears the baselines, and what I would measure next.

## A Transcript case studies

**Case 1 — the Putnam answer-set tautology (verbatim).** On `putnam_2018_a1` (characterize the pairs  $(a, b)$  with  $1/a + 1/b = 3/2018$ ) a model earns a comparator “pass” without proving anything, by redefining the answer set to be the question:

```
abbrev solution : Set (ℤ × ℤ) := {p | 1/p.1 + 1/p.2 = 3/2018}
theorem putnam_2018_a1 ...: (1/a + 1/b = 3/2018) ↔ ((a,b) ∈ solution) := by
rfl
```

The equivalence collapses to reflexivity; the six real solution pairs are never established and the statement is silently replaced. A second observed variant substitutes an outright wrong finite set. The guard infers the answer shape from the challenge and rejects both, which is why the agent “fails” Putnam on guarded runs — the guard is working, and no real proof was lost. (Sources: `outputs/baseline/*/putnam_2018_a1/solution.lean`.)

**Case 2 — same-model repair vs. cross-model handoff on the same problem.** The positive and negative findings share one exhibit. In R-G, `p10_factorial_pow` is closed when GPT-OSS receives its own failed candidate plus the exact Lean diagnostic (an unproven upper-bound obligation for  $n > k$ ) and repairs the induction — a same-model loop. In H-GQ, `p05_gcd_mersenne`, which GPT-OSS proves alone, is *lost* when GPT’s proposal is routed to Qwen: Qwen’s repair rewrites toward a different approach and abandons the near-closed trajectory. The verbatim exchanges (Lean message → model response → outcome) are committed as `transcript.json` under the corresponding validation-run problem directories; the working GPT proof of `p05` is at `outputs/baseline/openai/gpt-oss-120b/p05_gcd_mersenne/solution.lean`.

## B Failure taxonomy on `S_dev`

## C Spend ledger

Per-problem spend for the integrated-agent snapshot on `S_dev` (Table 5), and campaign totals. All figures are OpenRouter `usage.cost`; no problem approached the \$1 cap.

Campaign totals: S/R/H development matrix  $\approx$  USD 0.33; integrated-agent snapshot USD 0.41; each full16 judge-caps night  $\approx$  USD 1.7 (16 problems, \$1 max each); directed `p09` pilot  $\approx$  USD 0.25. Cumulative project spend stayed well inside the provided budget; the binding constraint on hard problems was wall-clock, not dollars.

**Table 4:** Substantive failures classified into the task’s four categories. “Formalization” = a correct informal idea that would not compile; “stuck repair” = a near-miss proof the models could not close under budget.

| Problem        | Dominant failure mode                                                                  |
|----------------|----------------------------------------------------------------------------------------|
| p09_imo1964    | Stuck repair (solvable by sampling scale; not under this run’s per-rung budget)        |
| rmo_2000_2     | Stuck repair (reachable by hand; models did not rediscover the cast-and-squeeze idiom) |
| rmo_2000_3     | Failed formalization / stuck repair (no accepted sub-proof)                            |
| putnam_2018_a1 | Correct target, but only tautological “proofs” produced; rejected by the guard         |

**Table 5:** Integrated agent, S\_dev snapshot (measurement caps \$0.20/1 h).

| Problem           | USD          | Wall (s)      |
|-------------------|--------------|---------------|
| p01_linear        | 0.000        | 119           |
| p03_sq_ge_two_ab  | 0.000        | 138           |
| p05_gcd_mersenne  | 0.000        | 105           |
| p06_pow_mod       | 0.004        | 196           |
| p10_factorial_pow | 0.009        | 387           |
| p09_imo1964       | 0.136        | 2111          |
| rmo_2000_2        | 0.102        | 3392          |
| rmo_2000_3        | 0.077        | 1948          |
| putnam_2018_a1    | 0.083        | 2062          |
| <b>Total</b>      | <b>0.411</b> | <b>10 458</b> |

## D Judge-caps stress: full16, two nights

Under the judge’s \$1/8 h caps, a parallel run over all sixteen sample problems passed **9/16 substantive** on two independent nights, the same nine each time (p01–p08, p10). The seven that resist are the hard tail (p09; rmo\_2000\_2/3/6, rmo\_2001\_2; putnam\_2018\_a1, putnam\_2020\_a2). Across all configurations p09 has been closed substantively (§6), a frontier of 10/16 on this suite. Both nights show the residual bank saving and reusing verified lemmas on the RMO problems without assembling the locked goal — the hollow-bank boundary in practice.

## E Run identifiers and artifacts

Every table above is backed by committed artifacts (result.json, transcript.json, events.jsonl). Load-bearing runs:

- S/R/H development matrix: registry Mx-\*Sdev, runs 20260826T\*Z, fixed-kit commit family 7baca49.
- Integrated-agent snapshot (S\_dev, 5/9): 20260830T010258Z under outputs/sdev-snapshot/.

- full16 judge-caps nights (9/16): 20260830T050459Z and 20260901T151104Z under `outputs/full16-*/`.
- Directed p09 pilot (PASS at `independent_slot_fallback`, empty bank): 20260831T061555Z.
- Partial validation campaign (three-seed agent replication; router baseline and full seeded matrix not completed) under `outputs/validation/`; `gate away/validation_gate_2026-09-02.md`.

## F Kit changes declared

I inject a small manifest of theorem/definition/numeric names into problem metadata so the acceptance guards can lock the problem shape, and I keep local, harder-locked copies of the two Putnam fixtures plus a corrected constant in one RMO fixture for lab use. The comparator and scoring logic are untouched; full diffs are in the repository README.

## G Repository map

The submission is a single agent module implementing the required interface. The development harness, the six Part Two arms, the repair core, the slot-combine and sampling machinery, the residual lemma bank, and the acceptance guards live alongside it, each in its own directory, with the human-readable experiment registry and split. The shipped agent is the readable ladder of §2; the wider inventory is development scaffolding, not the graded policy.

## H LLM tools used for development and writeup

Development and writeup used coding assistants and chat models (Claude Code / Codex-class) for scaffolding, debugging, and drafting. Runtime solving uses only the two mandated OpenRouter models. The same disclosure is provided in the submission form.